Supporting Information

Structure-Aware Row Scaling for Block-Structured Mixed-Integer Programs

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This document contains the 0.1%-MIPGap Gurobi distributional, diagnostic, and performance-profile views (Scenario 2 of the main text). They mirror the corresponding 1%-MIPGap figures of the main text; the quantitative evidence for the scenario is in the main text tables.

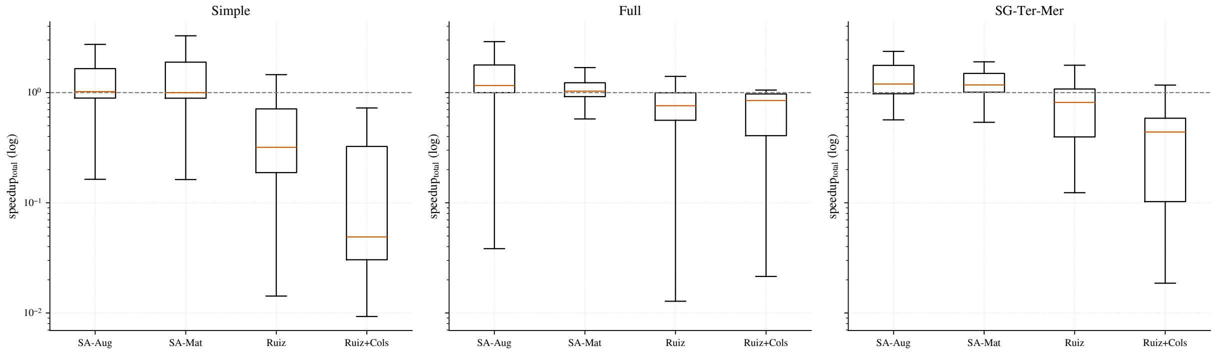
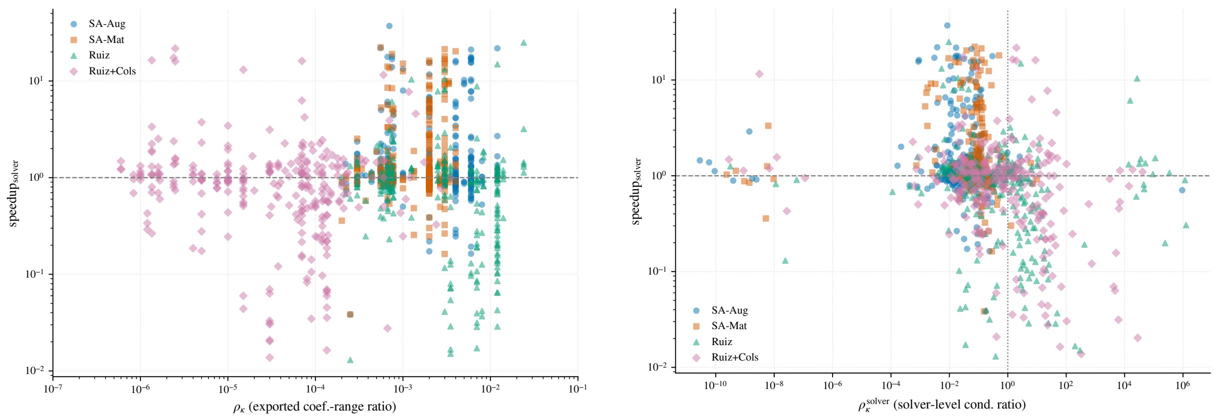


Figure S1: At 0.1% MIPGap, the Gurobi speedup distributions preserve the same qualitative pattern observed at the looser tolerance. Values above one indicate faster runs than the unscaled formulation.



(a) Static coefficient-range ratio. **(b)** Solver-level conditioning ratio.

Figure S2: At 0.1% MIPGap, the stricter Gurobi runs again separate static coefficient compression from solver-facing numerical behavior. The solver-level diagnostic remains more informative than the exported coefficient-range ratio.

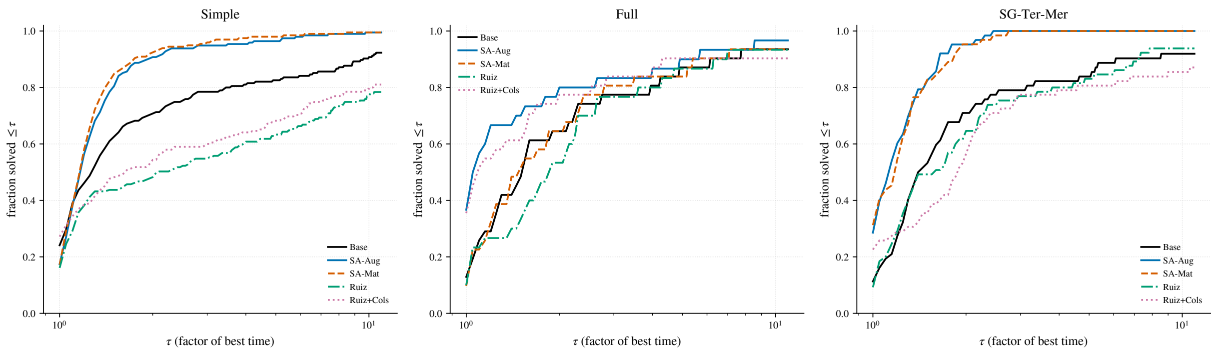


Figure S3: At 0.1% MIPGap, the Dolan–Moré profiles test whether the Gurobi speedups persist across the heterogeneous instance pool rather than only in averages.